G3XCAM1

Event Based Camera DataSheet

Product name : G3XCAM1
Model name : G3XCAM1

Rev.1.0.0

Preliminary

CenturyArks Co., Ltd.

Approval	*Responsible person*
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Revision History

Revision	Content	Date
Rev.1.0.0	Preliminary Release	2026.4.14

General description and application

G3XCAM1 consists of a PROPHESEE Event Based Vision Sensor (GENX320), a module that outputs MIPI and CPI (CMOS parallel interface) signals.
Please refer to the information site provided by PROPHESEE for the specifications of the sensor (GENX320) and related applications.
Please see the CenturyArks website for the latest information.
This product can be connected to the following OpenMV products:
OpenMV Cam H7 Plus, RT1062, N6 (planned)
For more information on OpenMV products, please see the following:
https://openmv.io/collections/cameras

Functions and Features

- GENX320(1/5” format, 320 x 320 array of 6.3µm pixels) PROPHESEE Event Based Vision Sensor
- Event data output with row-level 1µs-precision time stamping
- 0.05 lux Low light cutoff
- High dynamic range >120dB
- Multiple power modes:
- Embedded Anti-flicker filtering (AFK), Event-rate Controller (ERC), and Spatio-temporal Contrast filter (STC).
- Ambient light measurement
- ML-friendly compressed and uncompressed event data streams
- 1-lane MIPI D-Phy output interface
- Configurable 8-bit parallel output interface
- I²C interface.
- M12/M8 mount lenses are available. (Not included with the product.)

Referenced Documents

No.	Document Title
1	GENX320 DATASHEET (PROPHESEE)

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USE RESTRICTION NOTICE

This USE RESTRICTION NOTICE ("Notice") is for customers who are considering or currently using the event sensor camera products ("Products") set forth in this specifications book. CenturyArks Corporation (" CA ") may, at any time, modify this Notice which will be available to you in the latest specifications book for the Products. You should abide by the latest version of this Notice. distributor has its own use restriction notice on the Products, such a use restriction notice will additionally apply between you and the distributor. You should consult a sales representative of the distributor of on such a use restriction notice when you consider using the Products.

Use Restrictions

- The Products are intended for incorporation into such general electronic equipment as office products, communication products, measurement products, and home electronics products in accordance with the terms and conditions set forth in this specifications book and otherwise notified by CA from time to time.
- You should not use the Products for critical applications which may pose a life- or injury-threatening risk or are highly likely to cause significant property damage in the event of failure of the Products. You should consult your sales representative beforehand when you consider using the Products for such critical applications. In addition, you should not use the Products in weapon or military equipment.
- CA disclaims and does not assume any liability and damages arising out of misuse, improper use, modification, use of the Products for the above-mentioned critical applications, weapon and military equipment, or any deviation from the requirements set forth in this specifications book.
- CA is not responsible for any direct or indirect damage such as loss of stored data or damage to equipment caused by failure or use of this product.
- When installing and using this product, be sure to follow the warnings and precautions issued by the PC manufacturer and peripheral device manufacturer.
- This products and the application provided by CA do not guarantee the operation for all Computers.

Design for Safety

- CA is making continuous efforts to further improve the quality and reliability of the Products; however, failure of a certain percentage of the Products is inevitable. Therefore, you should take sufficient care to ensure the safe design of your products such as component redundancy, anti-conflagration features, and features to prevent mis-operation in order to avoid accidents resulting in injury or death, fire or other social damage as a result of such failure.

Export Control

- If the Products are controlled items under the export control laws or regulations of various countries, approval may be required for the export of the Products under the said laws or regulations. You should be responsible for compliance with the said laws or regulations.

No License Implied

- The technical information shown in this specifications book is for your reference purposes only. The availability of this specifications book shall not be construed as giving any indication that CA and its licensors will license any intellectual property rights in such information by any implication or otherwise. CA will not assume responsibility for any problems in connection with your use of such information or for any infringement of third-party rights due to the same. It is therefore your sole legal and financial responsibility to resolve any such problems and infringement.

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Governing Law

- This Notice shall be governed by and construed in accordance with the laws of Japan, without reference to principles of conflict of laws or choice of laws. All controversies and disputes arising out of or relating to this Notice shall be submitted to the exclusive jurisdiction of the Tokyo District Court in Japan as the court of first instance.

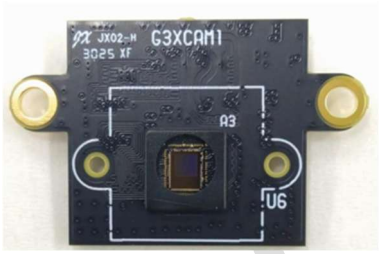
Other Applicable Terms and Conditions

- The terms and conditions in the CA additional specifications, which will be made available to you when you order the Products, shall also be applicable to your use of the Products as well as to this specifications book. You should review those terms and conditions when you consider purchasing and/or using the Products.

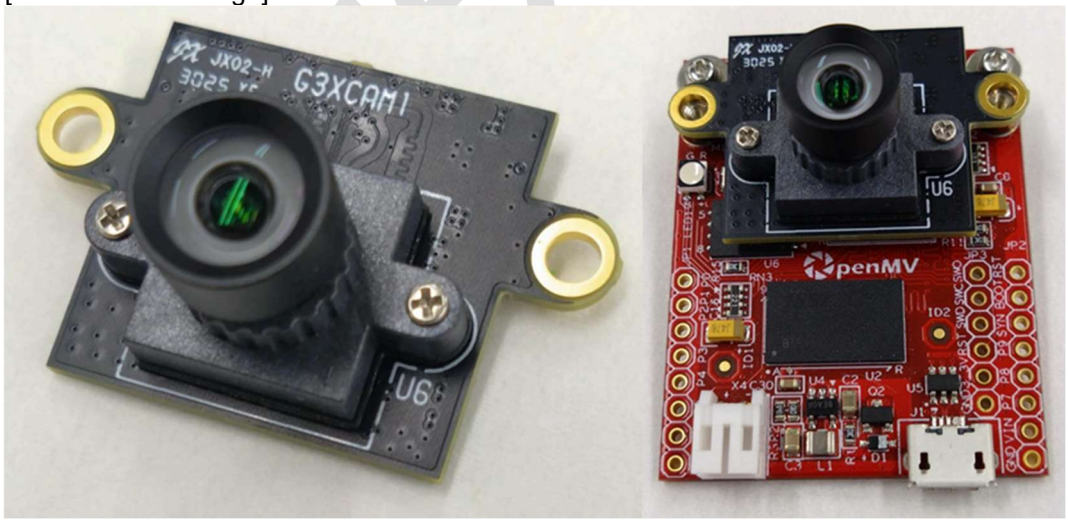
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1 Summary of Specifications

1.1. Table 1 List of Specifications

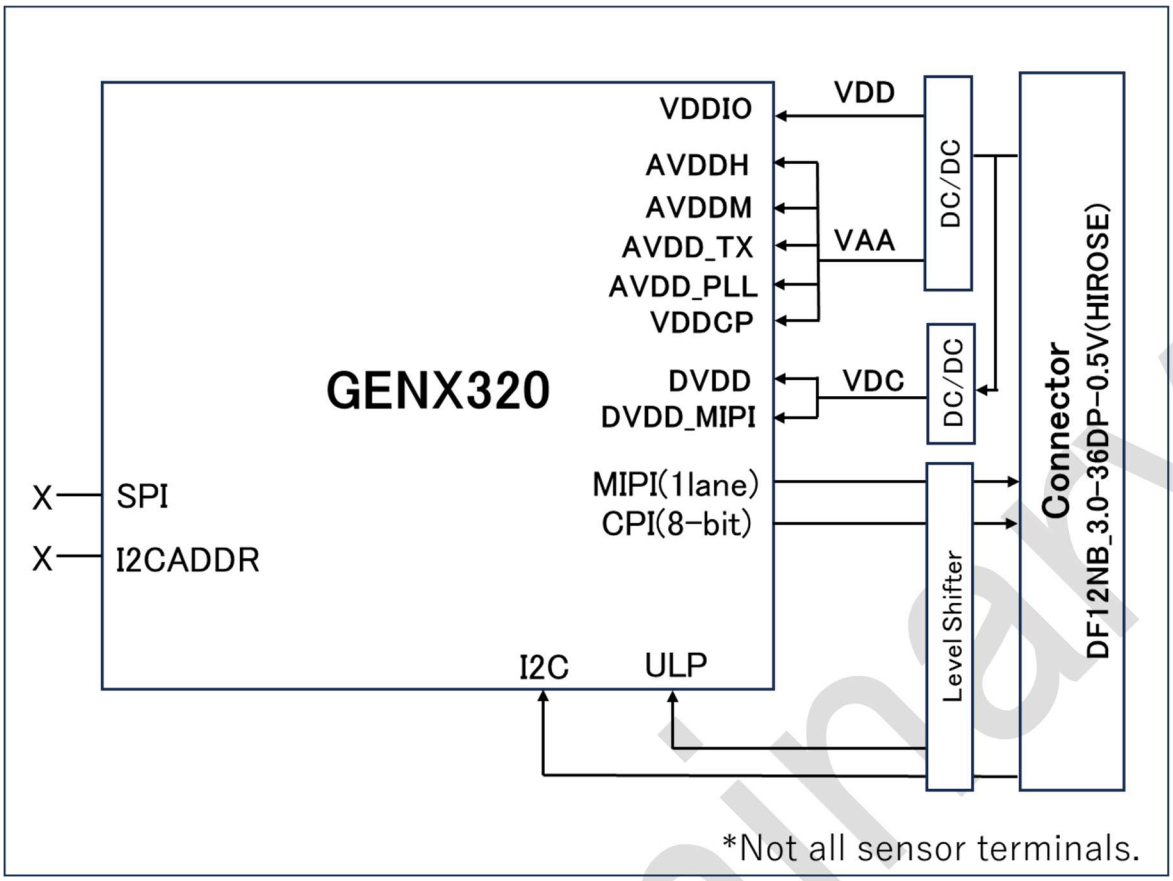
GENX320-CM2 Specifications			
Image Sensor	Image size	Type 1/5” (Diagonal 2.85 mm)	
	Number of effective pixels	320 (H) × 320 (V)	
	Pixel Size	6.3μm x 6.3μm	
Lens	M12/M8 mount lenses are available. (Not included with the product.)		
Output	MIPI CSI-2 (1 lane) or CPI(CMOS parallel interface)		
I/O	Input Clock (Typ)	10MHz	
	Control signal	I2C interface	
	Number of terminals	36 pins model: DF12NB(3.0)-36DP-0.5V (HIROSE)	
Others	Power supply	VDC	3.3 ± 0.15 V
	Power consumption	Multiple power modes	
	Operating temperature	T operation	–5～+55 ºC
	Storage temperature	T storage	–30～+80 ºC
	Recommended operating temperature	T specification	0～+50 ºC
	Operating humidity range	H operation	+80%RH or under, non-condensing
	Module dimensions	35.6mm x 22.5mm	
	Module weight	3.1 ± 0.1 g	

[Lens mounted image]



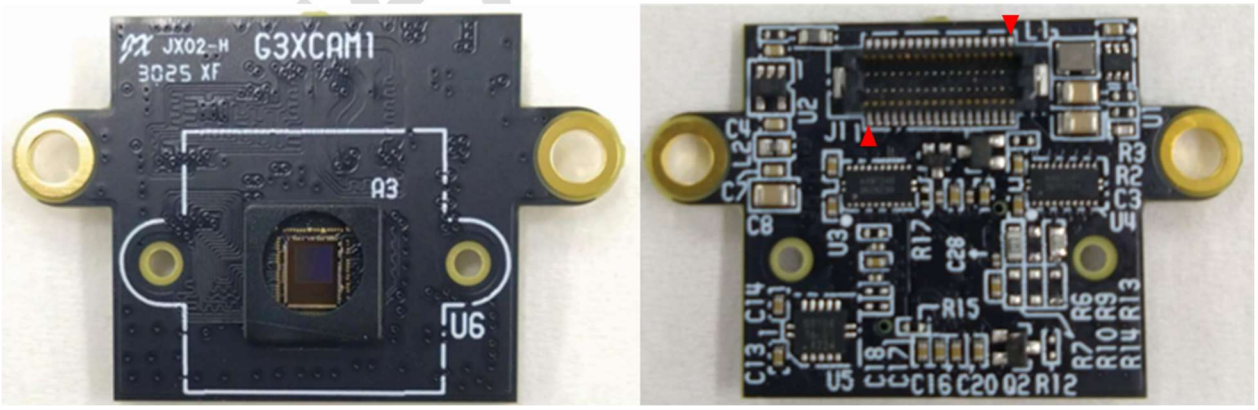
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1.2. Block Diagram



2 Pin Assignment & Description

2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35



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No	Symbol	I/O	A/D	Spec	Description
1	NC				Not Connect
2	VDC	Power	D	3.3V	
3	NC				Not Connect
4	NC				Not Connect
5	GND	GND	D	–	
6	MIPI_D0_P	O	D	3.3V	MIPI data plus
7	MIPI_CLK_P	O	D	3.3V	MIPI clock plus
8	MIPI_D0_N	O	D	3.3V	MIPI data minus
9	MIPI_CLK_N	O	D	3.3V	MIPI clock minus
10	VDC	Power	D	3.3V	
11	SPI_SSEL	I	D	3.3V	
12	DCMI_CLK	I	D	3.3V	
13	SPI_SCLK	I	D	3.3V	
14	DCMI_PCLK	I	D	3.3V	CMOMS I/F output clock
15	SPI_MISO	O	D	3.3V	
16	DCMI_VSYNC	O	D	3.3V	CMOMS I/F vertical sync
17	SPI_MOSI	I	D	3.3V	
18	DCMI_HSYNC	O	D	3.3V	CMOMS I/F horizontal sync
19	GND	GND	D	–	
20	VDC	Power	D	3.3V	
21	DCMI_RST	I	D	3.3V	Reset | active low
22	DCMI_D7	O	D	3.3V	CMOS I/F data 7
23	DCMI_FSIN	I	D	3.3V	
24	DCMI_D6	O	D	3.3V	CMOS I/F data 6
25	DCMI_PWDN	I	D	3.3V	
26	DCMI_D4	O	D	3.3V	CMOS I/F data 4
27	GND	GND	D	–	
28	DCMI_D3	O	D	3.3V	CMOS I/F data 3
29	DCMI_D1	O	D	3.3V	CMOS I/F data 1
30	VDC	Power	D	3.3V	
31	DCMI_D0	O	D	3.3V	CMOS I/F data 0
32	DCMI_SCL	I	D	3.3V	
33	DCMI_D5	O	D	3.3V	CMOS I/F data 4
34	DCMI_SDA	I/O	D	3.3V	
35	DCMI_D2	O	D	3.3V	CMOS I/F data 5
36	VDC	Power	D	3.3V	

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3 Electrical Characteristics

3.1. Absolute Maximum Ratings

Parameter	Symbol	target	Units
Absolute rating for 3.3V supplies	VDC	TBD	V
Storage temperature range	Ts	80	℃
Electrostatic discharge (ESD) **Human Body Model (HBM): JS-001-2014**	HBM	TBD	V
Electrostatic discharge (ESD) **Charged Device Model (CDM): JESD22-C101**	CDM	TBD	V

3.2. Recommended Operating Conditions

Parameter	Symbol	target	Units
Digital supply voltage	VDC	3.3±10%	V
Performance guarantee junction temperature	T operation	50	℃
Operating humidity range * non-condensing	H operation	+80 or under	%RH

4 Start Up Sequence

4.1. Power-on Sequence

Please refer GENX320 DATASHEET.

4.2. Power-Down Sequence

Please refer GENX320 DATASHEET.

5 Notes on Handling and Assembly

This lens module was designed to be assembled into electronic products. When using it, ensure safe design by heeding the following precautions.

5.1. Prevention of Electrostatic Discharge (ESD)

CMOS image sensors and other ICs can easily be damaged by static discharge. When handling the lens modules, take the following measures to prevent static discharge.

(1) Either use non-chargeable gloves, clothes or material. Also use conductive shoes.
(2) Use a wrist strap when handling directly.
(3) Install grounded conductive mats on the floor and working table to prevent the generation of static electricity.
(4) Ionized air is recommended for discharge when handling CMOS image sensors.
(5) For the shipment of mounted boards, use boxes treated for the prevention of static charges.

5.2. Notes on storage and operating environments

Do not store or use the lens modules in harsh environments with high temperatures, high humidity levels, and high concentrations of dust or in environments where condensation may form from moisture or dampness. It is also recommended that the lens modules not be stored for periods exceeding 1 month.

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6 Remodeling

Do not attempt to remodel or rework the lens module. Absolutely no guarantees will be given in the event that a module has been remodeled or reworked.

7 Safety standards

This lens module is a semi-finished product and, as such, no guarantees are given in regard to its compliance with any safety standards.
Take steps to ensure compliance with the applicable safety standards after it has been incorporated into the completed products. We can't guarantee it to any damage when incorporating in customer's product. Please evaluate sufficiently in customer's product.

8 Protection from dust and dirt

(1) Do not touch the lens surface with hand and make any objects to contact with it. When there is any dirt or other contaminant on the lens surface, remove it using a soft cloth or a similar object dipped in alcohol while taking care not to scratch or mark the surface.
(2) Keep in a dedicated case to protect from dust and dirt.

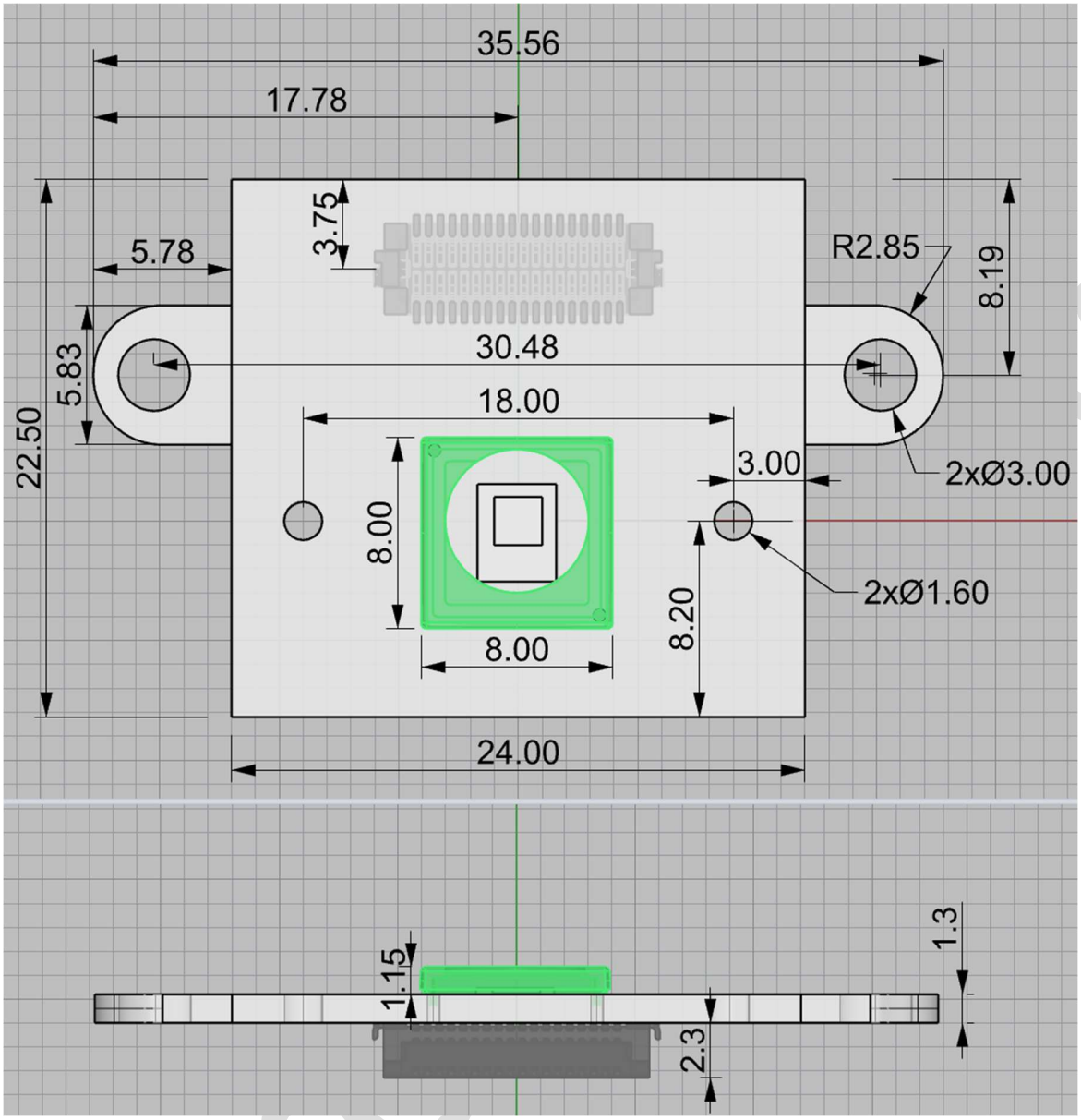
9 Notes on Assembly

1. How to secure the module in place

This product has electronic elements mounted on the back of the module. Please note the following points.
Depending on how the module is handled, electronic components may come off.
The material that the module in place causes an electrical short in the electronic elements on the back.

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10 Dimension



Proprietary

11 Notes on Handling and Assembly

When using this product, ensure safe design by heeding the following precautions.

1. Prevention of Electrostatic Discharge (ESD)

Event based sensors and other ICs can easily be damaged by static discharge. When handling these modules, take the following measures to prevent static discharge.

(1) Either use non-chargeable gloves, clothes or material. Also use conductive shoes.
(2) Use a wrist strap when handling directly.
(3) Install grounded conductive mats on the floor and working table to prevent the generation of static electricity.
(4) Ionized air is recommended for discharge when handling Event based sensors.
(5) For the shipment of camera, use bag or tray treated for the prevention of static charges.

2. Notes on storage and operating environments

Do not store or use the lens modules in harsh environments with high temperatures, high humidity levels, and high concentrations of dust or in environments where condensation may form from moisture or dampness.

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12 Packing specification

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